Growth performance of three strains of Nile Tilapia (*Oreochromis niloticus*) on four different feeds in Western and Central Kenya

by

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Abstract

A major impediment to recommending interventions to improve tilapia aquaculture in Africa is the lack of information on the performance of key inputs (feed and seed) in well characterized husbandry circumstances. The growth performance of three commercially available strains of Nile tilapia (*Oreochromis niloticus*) fed one of four different commercially available feeds during nursery and grow-out was assessed in Kenya. Each of the 12 treatments (three strains x four feeds) had four replicates (two in each of two ponds) at each of the two locations (Western and Central Kenya). The experiment tested starter feeds (phase 1, lasting 85 days) and grower feeds (phase 2, lasting 127 days). There were statistically significant differences in growth and FCR for fish grown on different starter and grower feeds within both areas, but not between fish strains at harvest in Central Kenya where growth of all strains was poor on all feeds. The mean weight over all strains at the end of the starter phase was 5.9 g for (FCR 2.4) and at final harvest 53.6 g (FCR 3.6). In the warmer environment in Western Kenya mean weight over all strains at the end of the starter phase was 19.5 g (FCR 2.2) and 189.8 g (FCR 1.9) at final harvest. The most cost effective combination was a high protein formulated feed and a feed efficient strain with 38% less average fixed feed costs than the next best combination. Mash, the cheapest feed tested, provided such poor weight gain that feed costs per unit gain were higher than for other feed strain combinations. Best production costs in Central Kenya (4.45 KES g^{-1}) were ten times that in Western Kenya (0.50 KES g^{-1}) and suggest tilapia farming in that region is unsustainable, although some niche operations may be feasible. Although performances of key inputs (feeds and strains) available in Kenya for tilapia aquaculture vary significantly, quality inputs are available in the country that can provide good outcomes in combination with sound management.

Introduction

There has been limited development of commercial aquaculture and continued underperformance of subsistence farming in Africa despite investments over a number of decades. In the case of Kenya the production of farmed fish grew from 1,000 MT in 2000 to 14,952 MT (2016) (Munguti et al., 2014a; KMFRI, 2017). One stimulus for this was the Fish Farming Enterprise Productivity Program (FFEPP), part of the Economic Stimulus Programme (ESP) launched in 2008/09. Through this the government intended to strengthen extension services and improve quality and distribution of key inputs such as seed and feed. More than 7,000 new fish ponds were built, growing the total area of fish farming from 220 ha to 468 ha. Farmed fish production peaked at 23,501 tonnes in 2013. A sharp drop (36%) in farmed fish production followed the end of the Stimulus Program and its associated subsidies, when many farmers stopped production (Obwanga et al., 2017). Sudden abandonment is common for interventionist approaches (Belton and Little, 2011) and underline the importance of sustainable and market-led business models in aquaculture.

There was also evidence that some of the ponds funded under the ESP did not perform as expected. In Makueni County more than 92.5% of ponds underperformed and more than 70% of ponds were abandoned (Musyoka and Mutia, 2016). Today, these performance issues remain, with confusion as to which combination of suboptimal inputs or management are responsible for underperformance in Kenya (Munguti et al., 2014) and the lack of robustness of aquaculture value chains (Obwanga et al. 2017). Understanding the reasons for poor performance is required to be able to identify and implement appropriate solutions. However, this is hampered by lack of data on the performance in well characterized circumstances so that appropriate interventions can be identified.

The aquaculture production capacity in Kenya is estimated at 11 million tonnes per year (Nyandat and Owiti, 2013), well above the peak production of approximately 24,000 tons in 2013. The commonest aquaculture systems in Kenya are earthen and lined ponds, followed by cages, dams, and tanks with tilapia species (mainly *O. niloticus*) accounting for 75 to 90 percent of farmed fish in Kenya (KMFRI, 2017; Munguti et al, 2014a). Kenya has diverse habitats from sea level to 5,200 meters and altitude is one of the key factors influencing average temperature differences of 11 °C (day or night) between low and high altitude areas (Rothuis et al., 2011). Water temperature is a critical factor affecting the growth of *O. niloticus* (El-Sayed et al., 1996).

With respect to key inputs, the most recent feed studies for *O. niloticus* in Kenya date back more than a decade when the growth and economic performances of Nile tilapia fed with brans (Lili et al. 2005) and locally formulated feeds, including pig pellets (Lili et al. 2006) were investigated. Research and development efforts have increased productivity of *O. niloticus* mono-sex fingerlings in Kenyan aquaculture to (Githukia et al., 2015). Some hatcheries have imported improved stains (e.g. GIFT) or YY-males to optimize tilapia production. Current studies of productivity and cost associated with different inputs available today are required to establish benchmarks for aquaculture production in different climatic areas in Kenya. This is needed to provide a basis for evidence-based recommendations for the future development of the Kenyan aquaculture sector.

The aim of the present study was to provide a first step towards that goal by scientifically evaluating the performance of three strains of Nile tilapia (*O. niloticus*) used in Kenyan farms today with currently available feeds using well documented rearing conditions. The experiment was carried out in earthen ponds, one of the most common farming systems in use in Kenya, in two of the main climatic zones to assess the performance of these inputs under a well-documented management regime. .

Materials and methods

Origin of stock

Three *O. niloticus* strains used were: strain 1, GMT® from the Fishgen Ltd. (UK, YY-technology) obtained from Jasa Fish Farm (Thika, Central Kenya); strain 2, Til Aqua Silver NMTTM of Til Aqua International BV (YY-technology), obtained from Jambo Fish (Mumias, Western Kenya); and strain 3, *O. niloticus,* a synthetic domesticated stock with fish derived from Lake Victoria, Lake Kyoga, and Lake Turkana obtained from Jewlet Enterprises, Kamwala Fish Hatchery (Kendu-Bay, Western Kenya).

Feed selection

Feeds from two local and two international feed manufacturers were used and obtained from commercial suppliers. The range of products included starter pellets, starter crumble, grower feeds, and mash. Mash sourced from Jewlet (40% crude protein, 85 KES kg^{-1}) was denoted feed 1. Starter pellets were sourced from Raanan (1mm, 48% crude protein, 300 KES kg^{-1}) denoted feed 2, starter crumble from Sigma Feeds Ltd (40 % crude protein, 150 KES kg^{-1}) denoted feed 3, and from Skretting (1mm, 57% crude protein, 390 KES kg^{-1}) denoted feed 4. For grow-out, Jewlet mash (40% crude protein, 85 KES kg^{-1}) – feed 1, Raanan grower (2mm, 40% crude protein, 150

KES kg^{-1}),- feed 2, Sigma grower (2mm, 30 % crude protein, 100 KES kg^{-1}) – feed 3 and Skretting grower (2mm, 35% crude protein, 145 KES kg^{-1}) –feed 4, were used. The selection of feeds and feed types represent a mix of widely used products in extensive and intensive farming with high availability in Kenya (pers. communication with Farm Africa), and includes products at the higher and lower price spectrum. For validation of ingredient information as provided by supplier, feeds were analyzed for dry matter, crude ash, crude protein, crude fat, and crude fiber at SoilCares; the results are summarized in Table 1.

Experiment locations

The trials were conducted at two sites with different environmental conditions. One site was located in Western Kenya, at the Lake Basin Development Authority (LBDA) near Kisumu, 1178 meters above sea level. The site has an annual average rainfall of 1321 mm, average yearly temperature of 22.9°C, with average monthly temperature highs ranging between 28 and 31°C and lows between 16 and 18°C (Climate-Data, 2018; NOAA, 2018). The second site was located at the University of Karatina, near Karatina Town in Central Kenya 1763 meters above sea level. This site has an average annual rainfall of 1449 mm, average yearly temperature of 17.5°C with average monthly temperature highs between 20 and 27°C and lows between 11 and 14°C.

While precipitation patterns are relatively similar in terms of absolute rainfall, the differences between average annual temperatures is 5.4°C. Kisumu has year-round high temperatures, with only 3°C difference in highs, while Karatina has generally lower temperatures between June and August (Climate-Data, 2018; NOAA, 2018). Monthly averages of the pond water temperatures taken daily (08:00 h and 17:00 h) throughout the experiment demonstrate the absolute and seasonal difference between the sites (Table 2).

Experimental set up

The experiment was carried out in two phases in order to assess productivity and production costs between starter and grower feeds. During the first phase of the experiment fingerlings were randomly selected and stocked at 100 fish per hapa and fed with starter feeds only (4800 fingerlings in total were stocked in 48 hapas in each of Western and Central Kenya). Fingerlings were nursed for 83.6 ± 1.0 (standard deviation) and 85.1 ± 1.3 days in Western and Central Kenya respectively. The second phase of the experiment was initiated once fish at one site had gained an average weight of 20 gram. At that time, after the standard length and weight for each fish was measured, stocking densities were reduced to 3 fish m^{-2} / 18 fish per hapa by random selection for the second phase of the experiment when grower feeds were fed to all fish at all sites. The entire trial was run over 212 days.

There were a total of 12 treatments (four feeds by three strains) and each treatment had four replicates resulting in each experimental site having 48 hapas. At each experimental site, three ponds (each approximately 250 m^2 water surface area and 1-meter depth) were set with 16 hapas (3m x 2m x 1m). Two replicates for each treatment was placed in each of two ponds.

Fingerlings were ordered at an average weight between two and three grams for each strain, but given transport over long distance, hatcheries supplied fish varying significantly in weight / size. In Central Kenya, fingerlings were of an average stocking weight of 1.12 ± 1.50 g overall with strain 1 of average 3.0 ± 0.98 g, strain 2 of 0.29 ± 0.27 g, and strain 3 of 0.08 ± 0.01 g. In Western

Kenya, the average stocking weight of fingerlings was 1.17 ± 0.70 g, with strain 1 of 1.05 ± 0.12 g, strain 2 of 2.03 ± 0.33 and strain 3 of 0.43 ± 0.12 g.

During phase one of the experiment, fingerlings were fed three times per day (09:00 h, 01:00 h, 16:00 h); during phase two, fish were fed twice a day (09:00 h and 16:00 h). The daily amount of feed was calculated based on the body weight of fish in grams: fish < 10 g = 7% of BW; fish > 10 < 20 g = 6 % BW; fish > 20 < 30 g 5% of = BW; fish > 30 < 50 g = 4% of BW; fish > 50 < 150 g = 3 % of BW; and fish > 150 g = 2% of BW. Sampling was done every seven days during phase one and every 14 days during phase two of the experiment. To reduce stress, fish was not fed on the day of sampling. The main purpose of sampling was to determine the weight of the fish and to adjust feeding. Growth was monitored by taking the individual weight of 30 randomly selected fingerling (phase one) from each hapa using electronic weighing balances (readability 0.01 g). During phase two, the number of randomly selected fish was reduced to nine fish per hapa. Water quality monitoring included 1) water temperature and DO recording (08:00 h and 17:00 h) and 2) Ammonium (NH4) (weekly). Measurements of DO and NH4 were primarily taken to inform management responses, if necessary, other than the weekly water exchange at 30% of water for each pond. The performances of the different strain and feed combinations were evaluated based on the weight gain (wg), the feed conversion ratio (FCR) [feed given (g)/body weight gain (g)], feed costs (KES??) and the average fixed costs [wg*FCR*feed costs/kg].

Data analysis

During the first phase, for each site, a fixed model effect was used to analyze the data. The analyzed traits were body weight at the end of phase one, feed conversion rate (FCR) and, calculated from these data and feed costs, the cost of production per kg of fish. The statistical model included nursery pond, fish strain, type of feed and the interaction between fish strain and type of feed as fixed effects. Average stocking weight per hapa nested within the nursery pond and the nursery period in days were included as covariates to account for the effect of differences in stocking weight, and growth period. During the second phase, the body weight at harvest (final body weight), FCR and cost of production per kg of fish of the second phase were analyzed using fixed model effect including grow-out pond, fish strain, type of feed, sex of fish and the two-way interactions of fish strain with type of feed and sex of fish. The average stocking weight at the beginning of phase two, nested within grow-out pond, was treated as a covariate to account for the effect of differences in stocking weight. The Bonferroni adjustment method was used to correct for the multiple pairwise comparisons. All analyses were done using SPSS (Statistical Package for the Social Sciences, Inc. 2011).

Results

Overall survival rates across individual replicates at each phase of the experiment were similar at the two locations: 85.0 % ± 10.4 (standard deviation) in Central and 84.0 % ± 13.7 in Western Kenya during the starter phase, and 94.2 % ± 6.3 in Central and 94.8 % ± 9.8 in Western Kenya during the grower phase. In neither phase were mortalities concentrated in any particular treatment or pond, but were spread throughout treatments and replicates.

In Central Kenya, over all strains and feeds, the average harvest weight of individual fish at the end of the starter phase was 5.9 ± 8.4 g compared with 19.5 ±13.7 g in Western Kenya. Thus the average stocking weight for the grower period was 7.1 g ± 8.3 g in Central Kenya and 21.7 g ± 11.7

g in Western Kenya, achieving average weights of 53.6 ± 43.0 g and 189.8 ±99.1 g and standard lengths of 10.6 ± 2.9 cm and 16.4 ± 2.7 cm, respectively at harvest.

There were significant effects of strain, feed (both starter and grower feeds) and pond on weight gain and FCR, and significant interactions between strain and feed in Western Kenya (Table 3). In contrast, there was no significant effect of strain on weight gain or FCR during the period fish were on grower feed in Central Kenya. Interactions between strain and feed were significant only for weight gain when fish were on starter feeds. There were no significant effects of pond on FCR in Central Kenya.

Sex could not be determined in the younger fish, and FCR was calculated for the total population in a hapa. Therefore sex and sex x strain effects could only be determined for weight gain on grower feeds and these effects were significant in both Central and Western Kenya. With respect to the co-variates, there was some variation for grow-out period in Central Kenya during the starter feed phase, and the effects of this on weight gain were significant (the fish growing for longer being larger). The variation in stocking weight of the fish had significant effects on weight gain (those stocked at larger size having a larger weight at harvest), but no significant effect on FCR, in both starter and grower periods in both Central and Western Kenya.

The magnitude of the differences of the effects of the main factors can be seen in the least squares means of weight gain and FCR for given strains, feeds or sexes (Table 4). The lack of significant differences among strains for weight gain and FCR on grower feeds in Central Kenya is clear as is their much lower weight gain, and higher FCR values, compared with those values for Western Kenya. The relative ranking of individual starter and grower feeds differed in Central and Western Kenya, but the differences in weight gain in Western Kenya were greater. Growth of males was significantly greater than females in both sites. However, given the significance of the interactions between fish strains and feeds, the main results are seen most clearly in a graphical representation of all strain and feed combinations (Figs 1-4) discussed in more detail below.

With respect to the starter phase, all strains had higher weight gain on feed 4 while their performances on the other feeds in Western Kenya were similar (Fig 1 top right graph). There was markedly lower weight gain for all strain and feed combinations in Central Kenya relative to Western Kenya. The three strains ranked consistently for FCR values, with strain 1 > strain 2 > strain 3, on all feeds in Central Kenya, and for feeds 2 and 3 in Western Kenya (Fig 1 bottom graphs). There was relatively little difference in FCR values between Central and Western Kenya.

In the grower phase (Fig 2), the lack of significant effects of strain on weight gain in Central Kenya was obvious, as was the much lower weight gain in Central Kenya compared with Western Kenya (Fig 2 top graphs). There was a trend of greater weight gain from feed 1 to feed 4 in Western Kenya, and of weight gain for strain 3 being greater than that of strains 1 and 2. There was a similar, but less obvious, trend of reducing FCR from feed 1 to feed 4 in Western Kenya. FCR values were high (around 4) and variable in Central Kenya, lower (around 2) and less variable in Western Kenya (Fig 2 bottom graphs).

Feed costs in both the starter and grower phases tended to reflect the same patterns as the respective FCR values in Figs 1 and 2 (Fig 3). The starter phase costs for strain 1 ≥ strain 2 > strain 3 for all feeds in Central Kenya and all feeds except feed 4 in Western Kenya (Fig 3 top graphs).

Costs for feed1 < feed3 < feed 2, < feed 4 in both Central and Western Kenya. In the grower phase, there was greater variability of costs for a given feed strain combination in Central Kenya relative to Western Kenya (Fig 3 bottom graphs). Feed 3 had the lowest cost over all strains in both Central and Western Kenya. Costs among strains varied with feed and location.

The critical information for farm management is the cost of the feeds relative to the production (ultimately profit) achieved and the average fixed cost (AFC) of rearing each strain and feed combination was calculated to assess this (Fig 4). In the starter phase AFC was far less in Western Kenya (40 KES g^{-1}) relative to Central Kenya (100 KES g^{-1}) over all strain and feed combinations, and lowest for feed 1 (<10 KES g^{-1}) in Western Kenya (Fig 4 top graphs). A trend was observed for higher AFC for feeds 2 and 4 reducing for feed 3 and being least for feed1 in both Central and Western Kenya. Higher AFC was observed for strains 1 and 2 for feeds 1 and 3 in Central Kenya and for feeds 1, 2 and 3 in Western Kenya. AFC was much less for the grower phase (<15 KES g^{-1} for most strain/feed combinations), and less in Western Kenya than in Central Kenya (Fig 4 bottom graphs). AFC for feed 1 > feed 2 > feed 4 > feed 3 for each strain in both Central and Western Kenya. A lower AFC for strain 3 was observed in Western Kenya for every feed but there was less consistency of strain performance in Central Kenya.

Discussion

The optimal conditions for good growth of tilapia are well known (Thomas and Michael, 1999; El-Sayed, 2006; Mjoun et al., 2010) and the loss in production, or yield gaps, when these are not used have been analyzed (Mengistu et al., 2019). Low water temperature constrains fish growth as fish can metabolically convert only a fraction of the nutrition provided in those conditions, and tilapia do not feed when temperatures drop below xx °C. The lower growth, higher FCRs (average

3.6) and limited differentiation in performance of feeds and strains in Central Kenya are clear indicators of culture in a sub-optimal environment. The lower temperatures there reduced growth and appear to have done so irrespective of the quality of feed or strain.

The more optimal environment in Western Kenya permitted a clearer differentiation of the growth potential of different strains and feeds. Although there were significant interactions between feed and strain some clear trends emerged. One of the imported formulated starter feeds (feed 4) had better performance over all strains than the other feeds and was differentiated in the ingredient analysis by higher crude protein and fat levels than the other feeds. Greater differentiation of feeds was observed after the grower phase. Higher weight gain was observed in the formulated feeds (feeds 2-4) compared with bran mash (feed 1) (Fig 2 top right graph). The higher performing of the formulated feeds, an imported (feed 4) and a locally produced (feed 3) feed were differentiated in the ingredient analysis by having higher crude fat levels than another imported feed (feed 2). The pattern of increasing weight gain from bran to formulated feed was mirrored in decreasing values of FCR indicating increasing efficiency of feed utilization using formulated diets. Higher weight gain and reduced FCRs were observed for strain 3 relative to the two other strains tested, on all feeds. Having identified these patterns of strain and feed differences in the more optimal environment of western Kenya it is pertinent to note that strain 3 showed significantly lower FCRs in the starter phase in Central Kenya. The pattern of FCR differences between feeds in the grower phase, although muted, paralleled that seen in Western Kenya.

However, the key to farm profitability is whether the additional production achieved by a better feed exceeds the additional cost of that feed. Feed costs differed significantly with bran being the cheapest, the formulated feeds more expensive with the imported feeds being most expensive. Costs were also affected by the efficiency of feed utilization. This was clearly seen in the close similarity in the pattern of costs with that of the FCRs in the starter phase (Fig 1 bottom left graph and Fig 3 top left graph) demonstrating the advantage of using more efficient fish strains. The analysis of the cost of production per unit weight of fish suggested that feed 1 (bran) provided the most cost-effective feed in the starter phase (Fig 4 top graphs). Two out of the three best performing combinations in Central Kenya were achieved using feed 1 with costs per g fish produced at the end of the starter phase ranging from xx KES g^{-1} for strain y to 5xx KES g^{-1} for strains x and z. Using feed 3 strain 2 showed xx% greater mean weight gain at xx% higher mean feed costs compared to strains x and x.

Two formulated feed (feeds 3 and 4) provided higher weight gains in the grower phase (Fig 2 top graphs) and were most cost-effective for a range of feed and strain combinations (Fig 4). The top three best performing combinations in Western Kenya were strain 3 + feed 4, strain 3 + feed 3, and strain1+feed 4 with average fixed costs of 0.50 KES g^{-1}, 0.67 KES g^{-1}, and 0.73 KES g^{-1}. The combination of feed efficient strain 3 with a comparatively expensive but efficient feed 4 provided a mean weight gain of 333.5 ± 11.1 g at mean feed costs of 165.5 ± 12.2 KES. This represents a 38% higher weight gain at 3% higher feed costs compared to the second best performing combination, strain 3 + feed 3, which achieved a mean weight gain of 240.9 ± 15.5 g at feed costs of 160.5 ± 12.9 KES 1 respectively. Feed 3 also provided the lowest average fixed costs of 4.45 KES g^{-1}, 5.17 KES $g^{-1}$and 5.55 KES g^{-1} with strain 2, strain 1 and strain 3 respectively during grow out in Central Kenya.

The present study did not test all feeds and strains available in Kenya, but the data contribute to a practical basis for recommendations to stakeholders in the Kenyan aquaculture sector. Overall, our study has shown that performances of key inputs (feeds and strains) available in Kenya for tilapia aquaculture vary significantly. First, they demonstrate that growth was poor and feed costs of production in the cooler environment in Central Kenya are almost ten times that of Western Kenya. They show the importance of culturing tilapia in environmental conditions within the optimum range in order to benefit from investment in more expensive but higher performing strains and feeds. Profitable aquaculture in the Central region may only be achieved in particularly favorable sites, stocking larger fingerlings to take account of shorter growing seasons, perhaps using heating or indoor phases when fingerlings are more sensitive to thermal conditions (Azaza et al., 2006, 2008). The proximity to large population centers in Nairobi and relatively high value markets is the attraction for farmers to the Central region and are sufficiently high to support greater production costs.

Second, they demonstrate the large differences in performance achieved by different feed and strain combinations grown in the same circumstances and that the relative outcomes can vary markedly between environments. These differences are not small. Even that between the top two best performing combinations in Western Kenya had a 38% difference in average fixed costs. Third, the use of high protein well-formulated feeds can lead to significant higher weight gain at comparatively lower feed costs, particularly when used in combination with feed efficient strains. Higher upfront investment in quality feeds and quality strains, especially during the grow-out period of fish can provide significant benefit. The use of mash the comparatively cheapest feed tested, produced lower weight gain in the grow-out phase and relatively high cost per unit weight of fish produced. Third, the work has demonstrated that there are quality inputs available in the country that can provide good outcomes in combination with sound management.

Successful farming will be dependent on well thought out business plans and sound operational management to obtain the efficiencies required. In Central Kenya, access to niche markets will be needed to achieve profitability. This business-like approach is not achieved by many farmers and promotion tilapia aquaculture will need to address these skills and take account of regional differences in environments, markets and differential access to quality inputs in promoting the industry.

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Table 1: Results on ingredient content analysis for sampled fish feeds

SAMPLE	Dry matter		Crude ash		Crude protein		Crude fat		Crude fibre	
	L	M	L	M	L	M	L	M	L	M
Starter										
Feed 1 mash	90	91.6	12	31.3	40.0	20.9	10.0	4.0	8.0	12.1
Feed 2 (3mm)	NA	92.1	8.0	7.9	30.0	33.6	5.0	7.3	6.0	4.9
Feed 3 (starter)	NA	93.3	NA	13.3	27.0	30.5	NA	5.3	NA	4.9
Feed 4 (starter)	NA	92.6	10.5	11.2	57.0	58.3	15.0	14.7	0.4	1.1
Grower										
Feed 1 mash	90	91.6	12	31.3	40.0	20.9	10.0	4.0	8.0	12.1
Feed 2 (4.5mm)	NA	91.0	8.0	8.3	30.0	35.5	5.0	6.9	6.0	5.4
Feed 3 (Grower)	NA	92.0	NA	12.2	27.0	33.8	NA	16.7	NA	4.7
Feed 4 (Grower)	NA	91.6	7.5	8.2	35.0	47.0	9.0	12.6	3.3	3.4

Values are expressed in percentages (%) per kilogram feed; L = as on label and M = measured in laboratory

Table 2: Average monthly water temperatures (°C) for Central and Western Kenya observed for the duration of the experiment.

Reading time	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb
Central Kenya								
0800h	18.4	19.2	19.1	20.2	19.4	20.6	20.5	20.8
1700h	19.9	21.6	22.0	24.3	22.5	23.9	24.3	25.5
Western Kenya								
0800h	26.8	26.8	27.5	27.9	28.4	28.1	27.1	27.2
1700h	27.1	27.5	27.9	28.3	29.1	28.5	27.4	27.5

Table 3: Significance levels for the effects of the factors, listed on the left of the table, given separately for the analyses of weight gain (WG) and feed conversion ratio (FCR) for starter and grower feeds in each of Central and Western Kenya. Statistically significant effects are given in bold. NA indicates tests were not possible or appropriate: sex could not be determined in the younger fish, and FCR was calculated for the total population in a hapa. Therefore sex and sex x strain effects could not be determined for starter feeds or for FCR. Significance of the effects of covariates are given at the bottom of the Table. The grow out period was identical for the tested groups in all cases except for starter feeds in Central Kenya, and therefore the test was only appropriate for that one circumstance.

	Central				Western			
	Starter		Grower		Starter		Grower	
Factors	WG	FCR	WG	FCR	WG	FCR	WG	FCR
Strain	**<0.01**	**<0.01**	0.20	0.89	**<0.01**	**0.05**	**<0.01**	**<0.01**
Feed	**<0.01**	**0.012**	**<0.01**	**0.05**	**<0.01**	**<0.01**	**<0.01**	**<0.01**
Pond	**<0.01**	0.24	**<0.01**	0.14	**<0.01**	**<0.01**	**<0.01**	**0.01**
Sex	NA	NA	**<0.01**	NA	NA	NA	**<0.01**	NA
Strain x Feed	**<0.01**	0.28	0.13	0.24	**<0.01**	**<0.01**	**<0.01**	**<0.01**
Strain x Sex	NA	NA	**<0.01**	NA	NA	NA	**0.02**	NA
Covariates								
Grow-out period	**0.01**	NA	NA	NA	NA	NA	NA	NA
Average stocking weight	**<0.01**	0.30	**<0.01**	0.35	**<0.01**	0.68	**<0.01**	0.09

Table 4. Least squares means (± SE) of the strain, feed and sex effects of weight gain (WG) and feed conversion ratio (FCR) for starter and grower feeds in each of Central and Western Kenya. Values within a column for a given group (strain, feed, or sex) with the same alphabetic superscripts are not significantly different but differ significantly from those with different alphabetic superscripts in that column.

	Central				Western			
	Starter		Grower		Starter		Grower	
Factors	WG	FCR	WG	FCR	WG	FCR	WG	FCR
Strain								
Strain 1	$7.1^{a} \pm 0.4$	$3.2^{b} \pm 0.3$	$45.6^{a} \pm 4.8$	$3.0^{a} \pm 1.3$	$21.0^{a} \pm 0.3$	$2.3^{a} \pm 0.1$	$162.9^{b} \pm 10.5$	$1.9^{a} \pm 0.0$
Strain 2	$6.0^{ab} \pm 0.3$	$2.4^{b} \pm 0.2$	$51.8^{a} \pm 2.7$	$3.8^{a} \pm 0.8$	$17.7^{b} \pm 0.9$	$2.2^{a} \pm 0.3$	$157.4^{b} \pm 9.5$	$2.8^{b} \pm 0.1$
Strain 3	$5.4^{b} \pm 0.2$	$1.7^{a} \pm 0.2$	$47.6^{a} \pm 3.0$	$4.0^{a} \pm 0.9$	$20.4^{a} \pm 0.8$	$2.0^{a} \pm 0.2$	$236.6^{a} \pm 11.4$	$1.8^{a} \pm 0.1$
Feed								
Feed 1	$4.9^{c} \pm 0.2$	$2.8^{b} \pm 0.3$	$33.4^{c} \pm 4.0$	$5.2^{b} \pm 0.6$	$18.6^{b} \pm 0.3$	$2.1^{ab} \pm 0.1$	$103.9^{d} \pm 8.5$	$2.7^{c} \pm 0.1$
Feed 2	$7.0^{b} \pm 0.2$	$2.3^{ab} \pm 0.2$	$45.5^{b} \pm 4.1$	$3.3^{ab} \pm 0.7$	$17.9^{b} \pm 0.3$	$2.1^{ab} \pm 0.1$	$172.5^{c} \pm 7.8$	$1.9^{b} \pm 0.1$
Feed 3	$4.6^{c} \pm 0.2$	$2.5^{ab} \pm 0.2$	$54.7^{a} \pm 4.1$	$2.8^{a} \pm 0.6$	$16.5^{c} \pm 0.3$	$2.5^{c} \pm 0.1$	$214.1^{b} \pm 10.1$	$1.6^{a} \pm 0.1$
Feed 4	$8.1^{a} \pm 0.2$	$2.1^{a} \pm 0.3$	$52.0^{ab} \pm 4.0$	$3.0^{ab} \pm 0.6$	$25.9^{a} \pm 0.3$	$2.0^{a} \pm 0.1$	$252.0^{a} \pm 7.9$	$1.5^{a} \pm 0.1$
Sex								
Female	NA	NA	$36.0^{b} \pm 2.4$	NA	NA	NA	$151.2^{b} \pm 11.3$	NA
Male	NA	NA	$60.7^{a} \pm 1.2$	NA	NA	NA	$220.1^{a} \pm 5.6$	NA

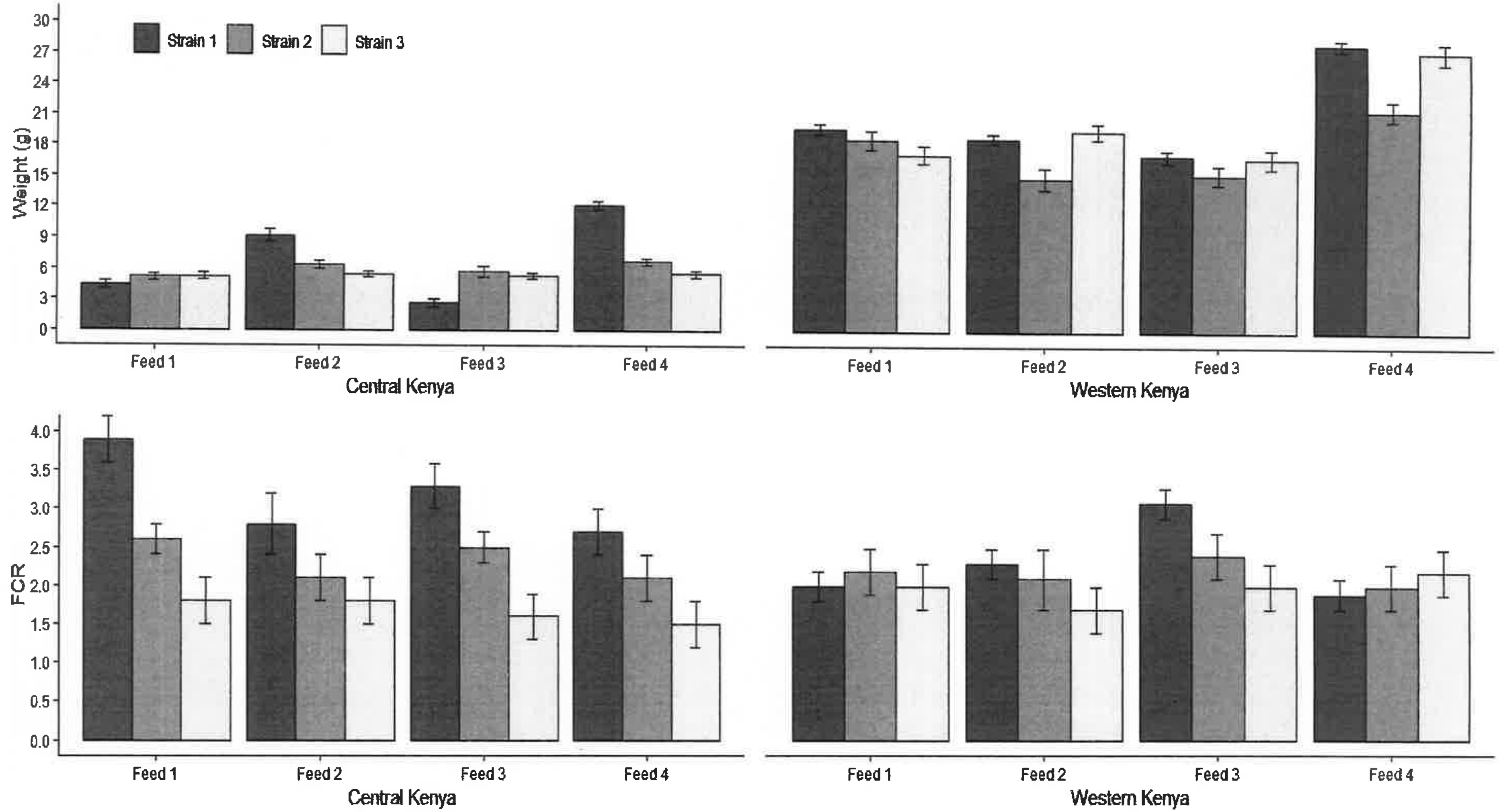

Fig. 1. Weight gain and FCR for starter feeds in Central and Western Kenya for each of the strain and feed combinations. The error bars are standard errors.

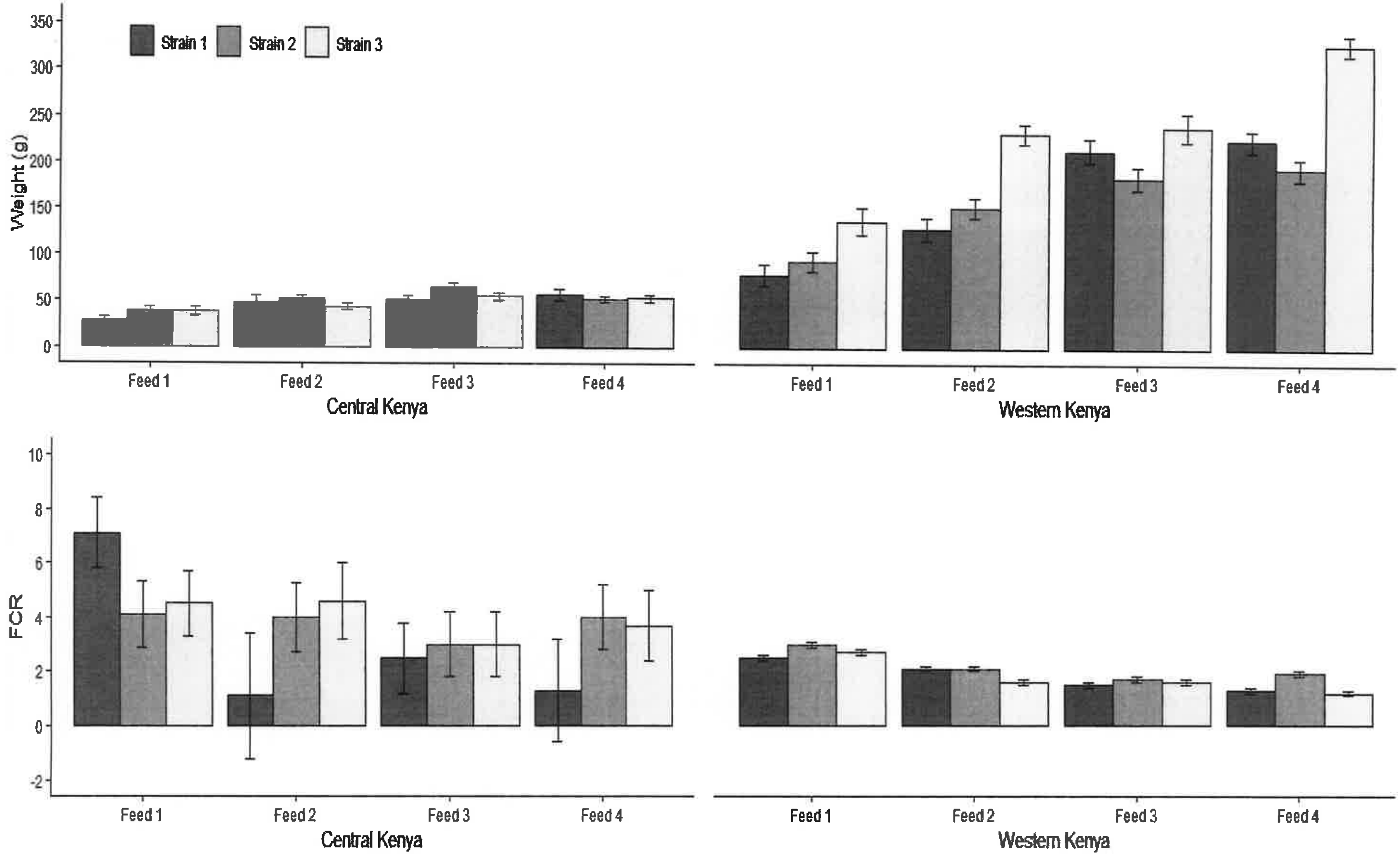

Fig. 2. Weight gain and FCR for grower feeds in Central and Western Kenya for each of the strain and feed combinations. The error bars are standard errors.

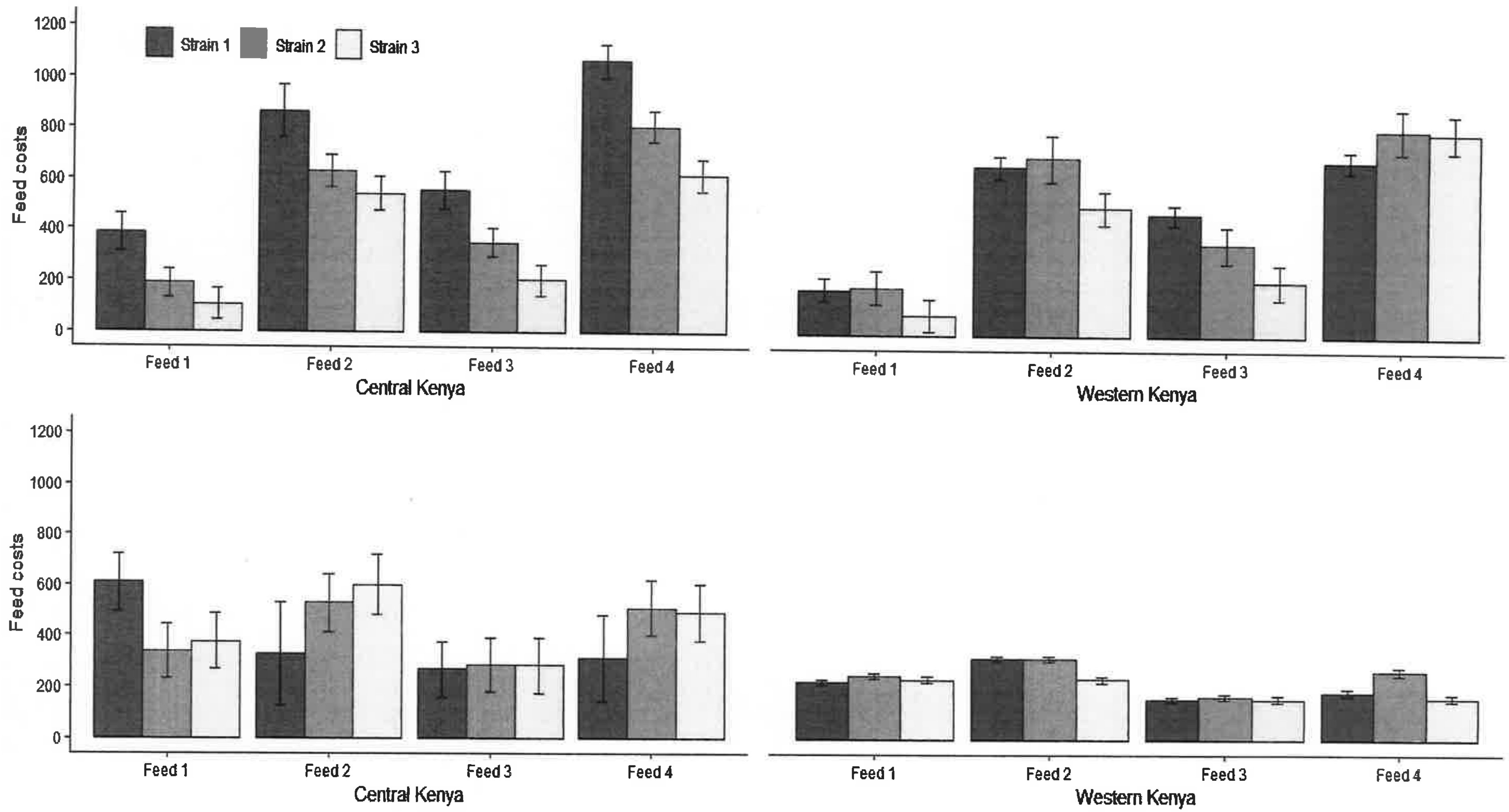

Fig. 3. Feed costs for starter feeds (top graphs) and grower feeds (lower graphs) in Central and Western Kenya for each of the strain and feed combinations. The error bars are standard errors.

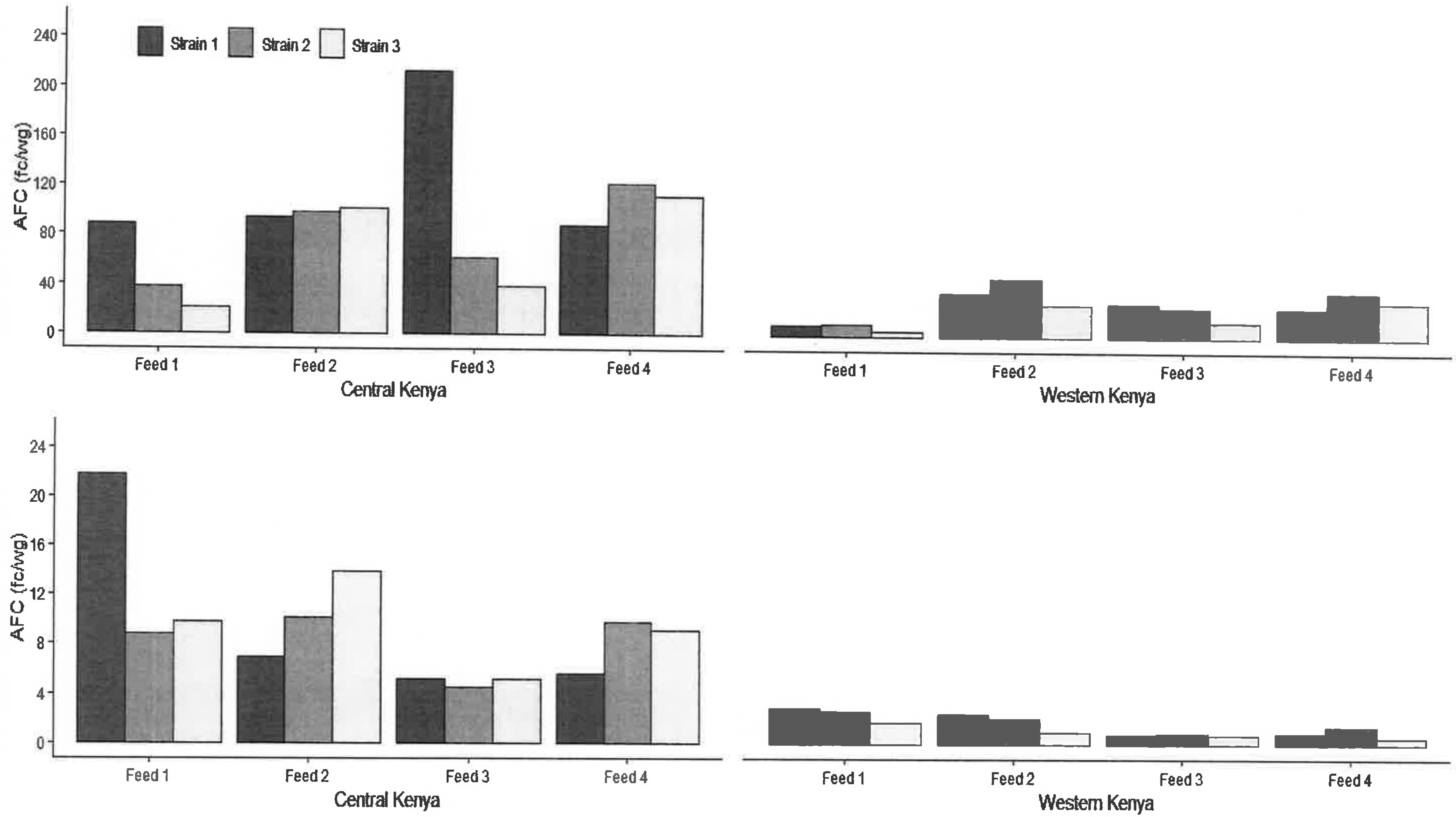

Fig. 4. Average fixed cost (AFC) for starter feeds (top graphs) and grower feeds (lower graphs) in Central and Western Kenya for each of the strain and feed combinations. There are no error bars since the data are a single set of calculations from the total of the values measured in the experiment.